

Marked variability of monacolin levels in commercial red yeast rice products: buyer beware!

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BACKGROUND: Red yeast rice (RYR) is a widely available dietary supplement used by millions of patients as an alternative therapy for hyperlipidemia. It contains 14 active compounds called monacolins that inhibit hepatic cholesterol synthesis. Although studies have suggested that some formulations of RYR may be effective and safe for lipid lowering, monacolin levels are not standardized among marketed products and are generally not published on labels. We evaluated monacolin levels in 12 commercial RYR formulations and tested for citrinin, a mycotoxin that is nephrotoxic in animals. METHODS: Each formulation of RYR was labeled "600 mg/capsule" of active product. Analyses for monacolins and citrinin were performed between August 2006 and June 2008 using high-performance liquid chromatography with mass spectroscopy-mass spectroscopy detection. Laboratory analyses of RYR products were conducted by ConsumerLab.com, White Plains, New York. RESULTS: There was marked variability in the 12 RYR products in total monacolins (0.31-11.15 mg/capsule), monacolin K (lovastatin) (0.10-10.09 mg/capsule), and monacolin KA (0.00-2.30 mg/capsule). Four products had elevated levels of citrinin. CONCLUSIONS: We found striking variability in monacolin content in 12 proprietary RYR products and the presence of citrinin in one-third of the formulations tested. Although RYR may have potential as an alternative lipid-lowering agent, our findings suggest the need for improved standardization of RYR products and product labeling. Until this occurs, physicians should be cautious in recommending RYR to their patients for the treatment of hyperlipidemia and primary and secondary prevention of cardiovascular disease.